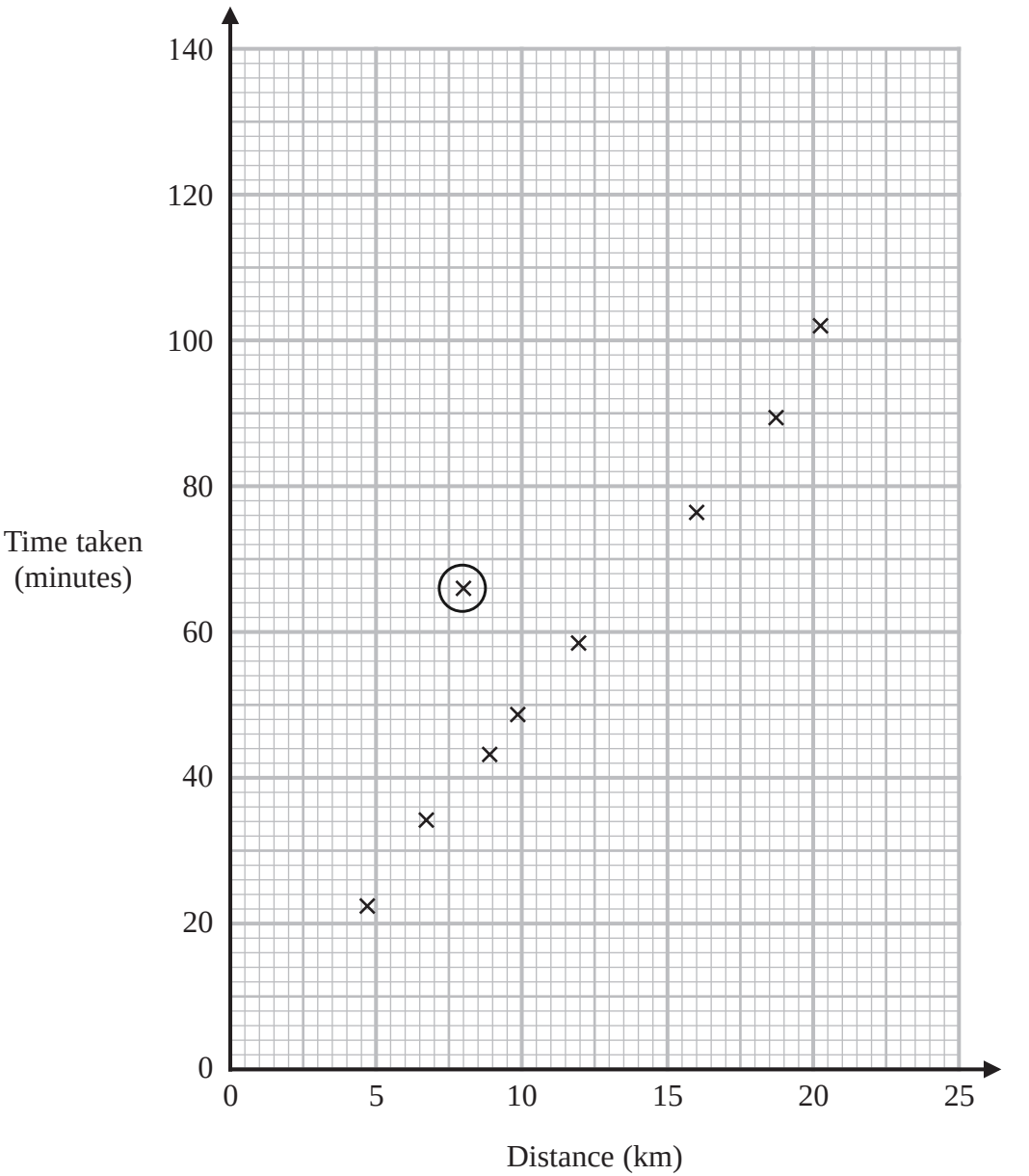
7 Giles is training to take part in a long distance run.
For each training session, he records the distance he ran and the time taken to run that distance.

Giles tries to run at the same speed in each session.

The scatter graph shows information about his results.



The point at 8 km is well above the general trend.

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On one of Giles' training sessions, he ran at a slower speed than on his other training sessions.

(a) (i) Explain how this can be seen from the scatter graph.

There is an outlier above the main trend:

he took much longer for that distance.

1 mark

More time for the distance means a slower speed.

(1)

(ii) On this training session, what distance did Giles run?

1 mark

Read the outlier's distance: 8 km.

8 km

(1)

Ignoring the training session in part (a), the equation of the line of best fit for the other eight results is

$y = 4.6x + 9$ where y minutes is the time taken for Giles to run x km.

(b) (i) Write down the gradient of the line of best fit.

1 mark

Gradient is the coefficient of x.

4.6

(1)

(ii) In the table below, put a tick (✓) in the box next to the sentence that describes what the gradient represents.

Giles runs 1 km in 9 minutes	
Giles runs 9 km in 4.6 minutes	
Giles runs 1 km in 4.6 minutes	✓
Giles runs 4.6 km in 9 minutes	
Giles runs 9 km in 1 minute	
Giles runs 4.6 km in 1 minute	

(1)

(Total for Question 7 is 4 marks)

1 mark

4.6 minutes for each extra kilometre.

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